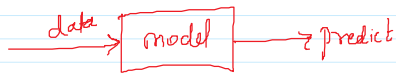


M.L → Subset of A.I

→ Build the model



→ Structured data is used in M.L

Dataset

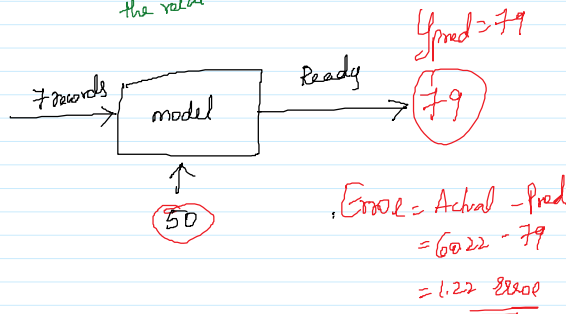
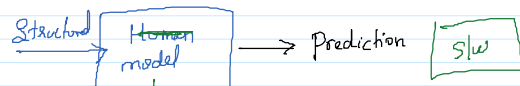
x	y
2	12.22
4	14.22
6	16.22
8	18.22
10	20.22
12	22.22
14	24.22
16	26.22
18	28.22

Train (75%)

Test (25%)

$y = x + 10.22$

(50) 60.22



70:30 → Train:Test

Index row	x ₁	x ₂	x ₃	x ₄	x ₅	x ₆	y
0							
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
11							
12							
13							
14							
15							
16							
17							
18							
19							
20							
21							
22							
23							
24							
25							
26							
27							
28							
29							
30							
31							
32							
33							
34							
35							
36							
37							
38							
39							
40							
41							
42							
43							
44							
45							
46							
47							
48							
49							

70 rec → row Index Selecting Randomly
0, 15, 2, 4, 15, 28, 91, 72, ...

30 rec X_{Test}
1, 3, 14, 19, 42, ...

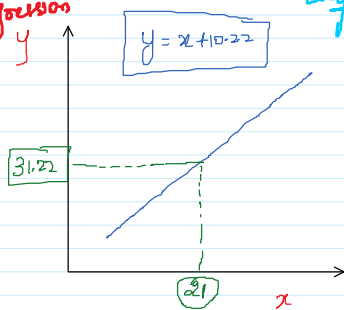
Y_{TRAIN}

Y_{Test}

Training data 99
= 0, 3, 5, 15, 24, 28, 29
no rec = 70

Test data
Unsupervised

Regression



x	y
19	32.22

1) Kind of data y var ⇒ Conti

Supervised

Classification

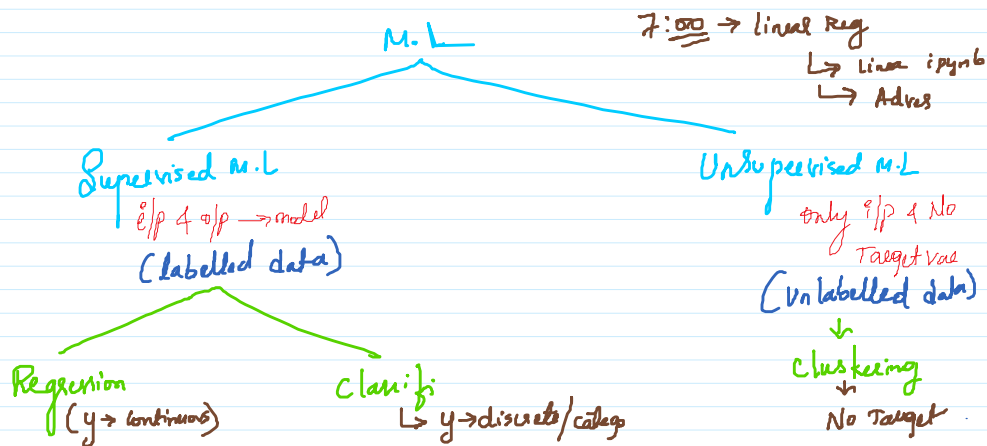
S	M	H	Result
90	80	75	F → 1
35	20	18	F → 0
90	81	83	P
25	18	33	F

Clustering

x ₁	x ₂	x ₃
9	3	2
200	170	160
8	4	1
202	170	177
9	32	846
170	160	177

< 11 > 100

* No Target



Linear Regression

- Supervised M.L
- labelled data
- $x \rightarrow$ features / ind
 $y \rightarrow$ Target var
- always x & y should have a linear Reg *
- there should not b/w i/p var (No Multicollinearity)

General eq: $y = m_1x_1 + m_2x_2 + \dots$

$(-2)^2 = 4$

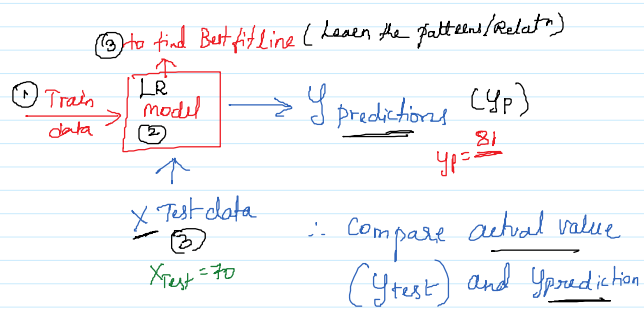
~~Residual~~ Residual Sum of Squares = $e_1^2 + e_2^2 + e_3^2 + e_4^2$

$= (y_{a1} - y_{p1})^2 + (y_{a2} - y_{p2})^2 + \dots$

$$RSS = \sum_{i=1}^n (y_{a_i} - y_{p_i})^2$$

Validation

actual	actual	
x	y	
10	10.22	Train: 180
5	15.22	
4	14.22	
70	80.22	Test: 20
150	110.22	



Error = $80.22 - 81 = -1.22$

Evaluate the model

R^2 Score

0.9 to 1 $\rightarrow$ Excellent
 0.7 to 0.9 $\rightarrow$ Good
 0.4 to 0.7 $\rightarrow$ Avg
 0.4 to 0 $\rightarrow$ Weak
 $< 0 \rightarrow$ Worst

$$\frac{S + M + Soc + H + L + L}{600}$$

% percentage
 $= 97\%$ FCD.

Logistic Regression Algorithm

- $\rightarrow$ Supervised M.L
- $\rightarrow$ labelled data
- $\rightarrow$ classification $\begin{cases} B_i \\ \text{Multi} \end{cases}$
- $\rightarrow$ principle: Probability Based
- $\rightarrow$ Target var $\rightarrow 0 \text{ \& } 1 \rightarrow B_i$

$\rightarrow$ Log + Regression

$$z = m_1x_1 + m_2x_2 + \dots + c$$

Eg:-

D			
Sub1	Sub2	Sub3	y
			Pass
			Fail
			Fail
			pass

Pass/Fail

Eg:- Percent = 85% Pass
 $= 30\%$ Fail

Cut-off 35%

Sub $> 35 \rightarrow$ Pass

Sub $< 35 \rightarrow$ Fail

Pass $\rightarrow$ class 0 $\rightarrow$ +ve class

Fail $\rightarrow$ class 1 $\rightarrow$ -ve class

Balanced dataset $\xrightarrow{\text{Yes/No}}$ 100 rec Yes
 100 rec No

Classification Unbalanced $\rightarrow$ 400 rec Yes class
 100 rec No class

Model is biased towards Yes class

Technique $\rightarrow$

② Compare Actual & Predicted
If it is same Define to True
If it is not same Define to False

③ Define the classes.
Women $\rightarrow$ true
Men $\rightarrow$ false

④ Consider only Prediction data
based on the Define class eg. If Pred W $\rightarrow$ true, M $\rightarrow$ false

Person ID	Actual Gender	Predicted Gender	Type
1	Woman	Woman	True Positive (TP)
2	Woman	Woman	TP
3	Woman	Woman	TP
4	Woman	Woman	TP
5	Woman	Woman	TP
6	Woman	Woman	TP
7	Woman	Man	False Negative (FN)
8	Woman	Man	FN
9	Woman	Man	FN
10	Woman	Man	FN
11	Man	Woman	False Positive (FP)
12	Man	Woman	FP
13	Man	Woman	FP
14	Man	Woman	FP
15	Man	Woman	FP

TP TN FP FN

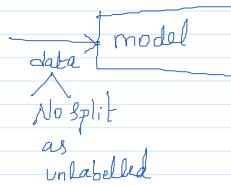
Actual Pred Type
Woman man FN
man woman TN
man woman FP
Woman woman TP

W $\rightarrow$ +
M $\rightarrow$ -

K-means algo

x_1, x_2, x_3

Model



Unsupervised

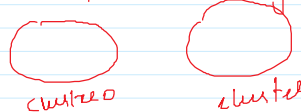
Y $\rightarrow$ Train
Y Test

x_1	x_2	x_3	Y
-	-	-	0
-	-	-	1
-	-	-	0
-	-	-	0
-	-	-	0
-	-	-	0
-	-	-	0
-	-	-	0

1) when target col is present? what to do?

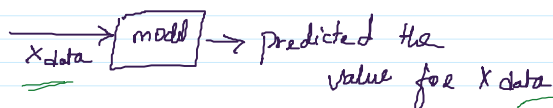
$\rightarrow$ remove/drop

2) Based on Y...? how many cluster?

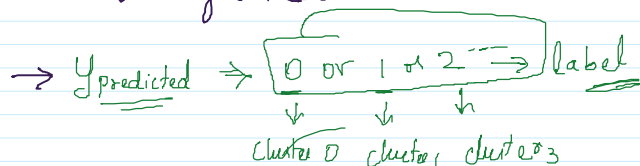


$\rightarrow$ we know the no of cluster $\Rightarrow$ Target V

① drop target var



x_1, x_2, x_3 Y actual
0
1
1
0
1
0
0



label $\rightarrow$ model predicted cluster/values
cluster 0

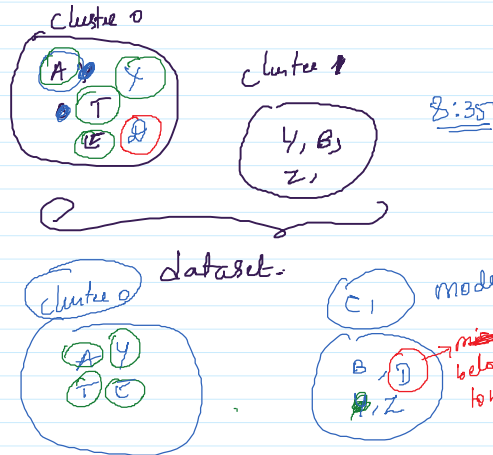
drop
 $\downarrow$ All
All
All
drop

drop

College name	X ₁	X ₂	X ₃	Y
A	85%	90%	92	0 ✓
X	55	50	65	1 ✓
Y	90	95	80	0
B	91	96	79	0
Z	-	-	-	-
D	-	-	-	-
E	-	-	-	-
T	-	-	-	-

drop

label → model predicted cluster/values



- 1) drop categories, Target
- 2) Train model by x data
- 3) pred - model generates a label (each row) 0, 1
- 4) Join the cat + Features + label

highest performance
cluster

College	X ₁	X ₂	X ₃	label	group
A	85%	90%	92%	0	group 1
B	-	-	-	1	group 2
C	-	-	-	1	group 2
D	-	-	-	0	group 1
E	-	-	-	1	group 2

K-means

→ Unsup ML

→ K → No of cluster

→ principle → Distance based algo / group based on similarities

$$E.O = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

- 1) Randomly we selected 2 Centroid
2 cluster
- K=2

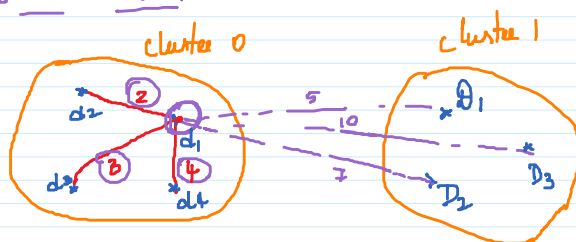
- 2) assign all pt to the cluster.

- 3) $C_{new} = (1, 5), (10, 20)$

- 4) assign all pt to the cluster based on new Centroids
- $C'_{new} \Rightarrow (6, 10), (60, 70)$

$$\begin{aligned}
 c'_{\text{new}} \quad c''_{\text{new}} &\Rightarrow ((6, 10)) \quad ((60, 70)) \quad \left. \vphantom{c'_{\text{new}} c''_{\text{new}}} \right\} \text{2nd } \downarrow \\
 \textcircled{5} \quad c''_{\text{new}} \quad c'''_{\text{new}} &\Rightarrow (6, 10) \quad (60, 70) \quad \left. \vphantom{c''_{\text{new}} c'''_{\text{new}}} \right\} \text{3rd NP change} \\
 &\quad \text{iterate steps}
 \end{aligned}$$

Evaluate the model

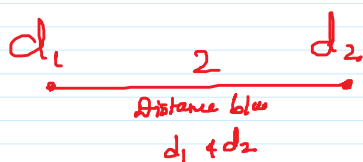


$$\begin{aligned}
 \text{Silhouette score} \\
 &= \frac{b-a}{\max(b,a)}
 \end{aligned}$$

$$a = \frac{2+3+4}{3} = 3.33$$

$$b = \frac{5+10+7}{3} = 5.12$$

$$\begin{aligned}
 S(d_1) &= \frac{5.12 - 3.33}{\max(3.33, 5.12)} = \frac{5.12 - 3.33}{5.12} = \underline{\hspace{2cm}} \\
 S(d_2) &
 \end{aligned}$$

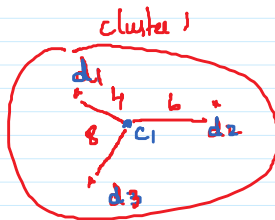
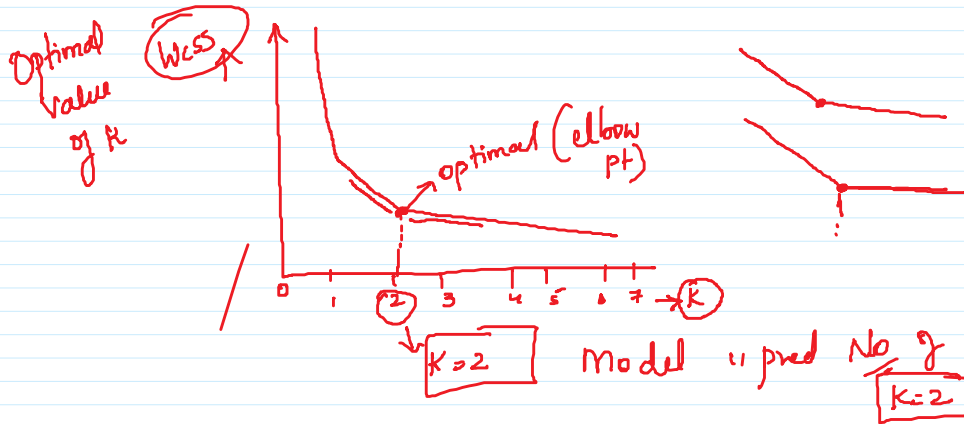


Assume if don't No of cluster

$$\begin{aligned}
 \min \quad & \boxed{k=2} \quad \underline{WCSS_2} \\
 & \boxed{k=3} \quad \underline{WCSS_3} \\
 & \boxed{k=4} \quad \underline{WCSS_4} \\
 & \boxed{k=5}
 \end{aligned}$$

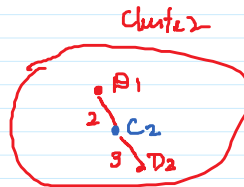
① step 1 $\rightarrow$ Random

Within cluster
Sum of Squares

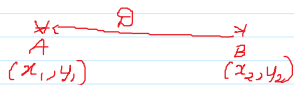


$$WCSS_1 = (4)^2 + (6)^2 + (8)^2$$

$$=$$



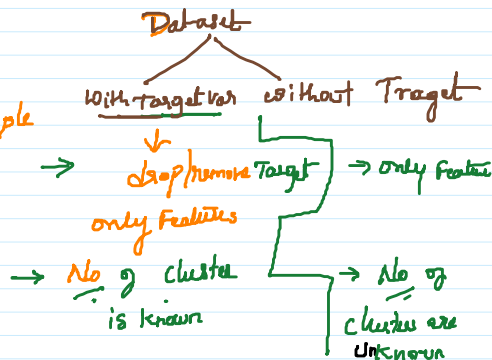
$$WCSS_2 = 2^2 + 3^2$$



$$Distance = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

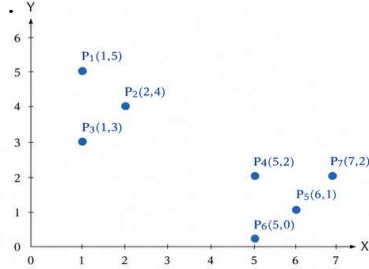
K Means Alg.

- Unsupervised M.L
- Unlabelled
- Distance Based algo → Principle
- k - No. of cluster
- Means → Average
- Numerical data



We need find
Optimal value of
K

K-MEANS CLUSTERING (K = 2) WITH 7 DATA POINTS



DATA POINTS		
Point	X	Y
P1	1	5
P2	2	4
P3	1	3
P4	5	2
P5	6	1
P6	5	0
P7	7	2

STEP 3: RECOMPUTE CENTROIDS (Iteration 1)

Cluster C1 has points: P1, P2, P3
 $C1_{new} = \left(\frac{1+2+1}{3}, \frac{5+4+3}{3} \right) = \left(\frac{4}{3}, 4 \right) = (1.33, 4.00)$

STEP 1: INITIALIZE CENTROIDS (CHOOSE K = 2)

Let $C_1 = P_1(1,5)$ and $C_2 = P_5(6,1)$

STEP 2: ASSIGN EACH POINT TO NEAREST CENTROID (Iteration 1)

Point	Distance to $C_1(1,5)$		Distance to $C_2(6,1)$		Assigned Cluster
	$\sqrt{(x-1)^2 + (y-5)^2}$	Value	$\sqrt{(x-6)^2 + (y-1)^2}$	Value	
P1	$\sqrt{(1-1)^2 + (5-5)^2}$	0.00	$\sqrt{(1-6)^2 + (5-1)^2}$	6.40	C1
P2	$\sqrt{(2-1)^2 + (4-5)^2}$	1.41	$\sqrt{(2-6)^2 + (4-1)^2}$	5.00	C1
P3	$\sqrt{(1-1)^2 + (3-5)^2}$	2.00	$\sqrt{(1-6)^2 + (3-1)^2}$	5.39	C1
P4	$\sqrt{(5-1)^2 + (2-5)^2}$	5.00	$\sqrt{(5-6)^2 + (2-1)^2}$	1.41	C2
P5	$\sqrt{(6-1)^2 + (1-5)^2}$	6.40	$\sqrt{(6-6)^2 + (1-1)^2}$	0.00	C2
P6	$\sqrt{(5-1)^2 + (0-5)^2}$	6.40	$\sqrt{(5-6)^2 + (0-1)^2}$	1.41	C2
P7	$\sqrt{(7-1)^2 + (2-5)^2}$	6.71	$\sqrt{(7-6)^2 + (2-1)^2}$	1.41	C2

Point	Distance to $C1(1.33, 4.00)$		Distance to $C2(5.75, 1.25)$		Assigned Cluster
	$\sqrt{(x-1.33)^2 + (y-4.00)^2}$	Value	$\sqrt{(x-5.75)^2 + (y-1.25)^2}$	Value	
P1	$\sqrt{(1-1.33)^2 + (5-4)^2}$	1.05	$\sqrt{(1-5.75)^2 + (5-1.25)^2}$	6.05	C1
P2	$\sqrt{(2-1.33)^2 + (4-4)^2}$	0.67	$\sqrt{(2-5.75)^2 + (4-1.25)^2}$	4.65	C1
P3	$\sqrt{(1-1.33)^2 + (3-4)^2}$	1.05	$\sqrt{(1-5.75)^2 + (3-1.25)^2}$	5.05	C1
P4	$\sqrt{(5-1.33)^2 + (2-4)^2}$	4.18	$\sqrt{(5-5.75)^2 + (2-1.25)^2}$	1.06	C2
P5	$\sqrt{(6-1.33)^2 + (1-4)^2}$	5.57	$\sqrt{(6-5.75)^2 + (1-1.25)^2}$	0.35	C2
P6	$\sqrt{(5-1.33)^2 + (0-4)^2}$	5.21	$\sqrt{(5-5.75)^2 + (0-1.25)^2}$	1.46	C2
P7	$\sqrt{(7-1.33)^2 + (2-4)^2}$	6.02	$\sqrt{(7-5.75)^2 + (2-1.25)^2}$	1.46	C2

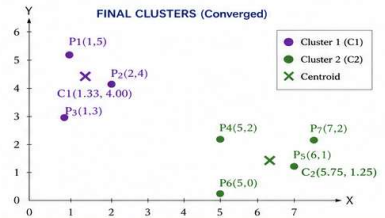
STEP 5: RECOMPUTE CENTROIDS (Iteration 2)

Cluster C1: P1, P2, P3
 $C1_{new} = (1.33, 4.00)$ (no change)
 Cluster C2: P4, P5, P6, P7
 $C2_{new} = (5.75, 1.25)$ (no change)

Centroids do not change $\Rightarrow$ Converged!

FINAL CLUSTERS

C1 (Center = (1.33, 4.00)) : P1, P2, P3
 C2 (Center = (5.75, 1.25)) : P4, P5, P6, P7



SILHOUETTE SCORE

Measures how well each point lies within its cluster.

For a point i :

- $a(i)$ = average distance to all other points in the same cluster
- $b(i)$ = minimum average distance to points in the other cluster
- Silhouette $s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$

$-1 \leq s(i) \leq 1$

- Close to 1 $\rightarrow$ well clustered
- Close to 0 $\rightarrow$ on the boundary
- Close to -1 $\rightarrow$ likely misclustered

Point	Cluster	$a(i)$ (avg intra-cluster distance)	$b(i)$ (avg distance to other cluster)	$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$
P1 (1,5)	C1	1.138	5.450	0.791
P2 (2,4)	C1	0.943	4.842	0.805
P3 (1,3)	C1	1.138	5.214	0.781
P4 (5,2)	C2	1.305	4.279	0.695
P5 (6,1)	C2	0.778	4.310	0.819
P6 (5,0)	C2	1.195	4.538	0.736
P7 (7,2)	C2	1.305	4.685	0.722

Average Silhouette Score

$$= \frac{(0.791 + 0.805 + 0.781 + 0.695 + 0.819 + 0.736 + 0.722)}{7} = \frac{5.349}{7} = 0.764$$

Interpretation

An average silhouette score of 0.764 indicates well-separated and compact clusters.

I $C_1 \rightarrow P_1, P_2, P_3$
 $C_2 \rightarrow P_4, P_5, P_6, P_7$

$C1_{new} (1.33, 4)$ $C2_{new} (5, 1.67)$

II $C1_{new} \rightarrow P_1, P_5, P_7, P_6$
 $C2_{new} \rightarrow P_2, P_3, P_4$

$C1_{new} (15, 17.21)$ $C2_{new} (65, 121.67)$

III $C1_{new} (15, 17.2)$ $C2_{new} (65, 121.6)$

same? no ---

same / No change in
Centroid, stops the
iteration

